



Top 48 CAT Venn Diagrams Questions With Video Solutions

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Questions

Instructions

For the following questions answer them individually

Question 1

Choose the set in which the statements are most logically related.

- A. No fishes breathe through lungs.
- B. All fishes have scales.
- C. Some fishes breed up stream.
- D. All whales breathe through lungs.
- E. No whales are fishes.
- F. All whales are mammals.

- A ABC
- B BCD
- C ADE
- D DEF

 [VIEW SOLUTION](#)

Question 2

Choose the set in which the statements are most logically related.

- A. All roses are fragrant.
- B. All roses are majestic.
- C. All roses are plants.
- D. All roses need air.
- E. All plants need air.
- F. All plants need water.

- A CED
- B ACB
- C BDC
- D CFE

 [VIEW SOLUTION](#)

Question 3

Choose the set in which the statements are most logically related.

- A. No lunatics are fit to serve on a jury.
- B. Everyone who is sane can do logic.
- C. None of your sons can do logic.
- D. Some who can do logic are fit to serve on a jury.
- E. All who can do logic are fit to serve on a jury.
- F. Everyone who is sane is fit to serve on a jury.

- A BDE

- B BEF
- C BDF
- D ADE

[VIEW SOLUTION](#)



Question 4

Choose the set in which the statements are most logically related.

- A. All candid men are persons who acknowledge merit in a rival.
- B. Some learned men are very candid.
- C. Some learned men are not persons who acknowledge merit in a rival.
- D. Some learned men are persons who are very candid.
- E. Some learned men are not candid.
- F. Some persons who recognize merit in a rival are learned.

- A ABE
- B ACF
- C ADE
- D BAF

[VIEW SOLUTION](#)

Question 5

Choose the set in which the statements are most logically related.

- A. All those who achieve great ends are happy.
- B. All young people are happy.
- C. All young people achieve great ends.
- D. No young people achieve great ends.
- E. No young people are happy.
- F. Some young people are happy.

- A ADE
- B ABF
- C ACB
- D ADF

[VIEW SOLUTION](#)

Question 6

Choose the set in which the statements are most logically related.

- A. Some beliefs are uncertain.
- B. Nothing uncertain is worth dying for.

- C. Some belief is worth dying for.
- D. All beliefs are uncertain.
- E. Some beliefs are certain.
- F. No belief is worth dying for.

- A ABF
- B BCD
- C BEF
- D BDF

[VIEW SOLUTION](#)



Question 7

Choose the set in which the statements are most logically related.

- A. Some of my closest friends disapprove of me.
- B. Some of my closest friends are aardvarks.
- C. All of my closest friends disapprove of me.
- D. All who disapprove of me are aardvarks.
- E. Some who disapprove of me are aardvarks.
- F. Some of my closest friends are no aardvarks.

- A BCD
- B ABD
- C BCE
- D ABE

[VIEW SOLUTION](#)

Question 8

Choose the set in which the statements are most logically related.

- A. First-year students of this college like to enter for the prize.
- B. All students of this college rank as University students.
- C. First-year students of this college are entitled to enter for the prize.
- D. Some who rank as University students are First-year students.
- E. All University students are eligible to enter for the prize.
- F. All those who like to are entitled to enter for the prize.

- A AEF
- B ABC
- C BEC
- D CDF

 [VIEW SOLUTION](#)

Question 9

Choose the set in which the statements are most logically related.

- A. Some mammals are carnivores.
- B. All whales are mammals.
- C. All whales are aquatic animals.
- D. All whales are carnivores.
- E. Some aquatic animals are mammals.
- F. Some mammals are whales.

A ADF

B ABC

C AEF

D BCE

 [VIEW SOLUTION](#)



CAT Previous Papers

with detailed video solutions and analysis



Question 10

Choose the set in which the statements are most logically related.

- A. All men are men of scientific ability.
- B. Some women are women of scientific ability.
- C. Some men are men of artistic genius.
- D. Some men and women are of scientific ability.
- E. All men of artistic genius are men of scientific ability.
- F. Some women of artistic genius are women of scientific ability.

A ACD

B ACE

C DEF

D ABC

 [VIEW SOLUTION](#)

Instructions

Applicants for the doctoral programmes of Ambi Institute of Engineering (AIE) and Bambi Institute of Engineering (BIE) have to appear for a Common Entrance Test (CET). The test has three sections: Physics (P), Chemistry (C), and Maths (M). Among those appearing for CET, those at or above the 80th percentile in at least two sections, and at or above the 90th percentile overall, are selected for Advanced Entrance Test (AET) conducted by AIE. AET is used by AIE for final selection.

For the 200 candidates who are at or above the 90th percentile overall based on CET, the following are known about their performance in CET:

1. No one is below the 80th percentile in all 3 sections.

2. 150 are at or above the 80th percentile in exactly two sections.
3. The number of candidates at or above the 80th percentile only in P is the same as the number of candidates at or above the 80th percentile only in C. The same is the number of candidates at or above the 80th percentile only in M.
4. Number of candidates below 80th percentile in P: Number of candidates below 80th percentile in C: Number of candidates below 80th percentile in M = 4:2:1.

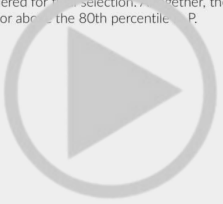
BIE uses a different process for selection. If any candidate is appearing in the AET by AIE, BIE considers their AET score for final selection provided the candidate is at or above the 80th percentile in P. Any other candidate at or above the 80th percentile in P in CET, but who is not eligible for the AET, is required to appear in a separate test to be conducted by BIE for being considered for final selection. Altogether, there are 400 candidates this year who are at or above the 80th percentile in P.

Question 11

If the number of candidates who are at or above the 90th percentile overall and also at or above the 80th percentile in all three sections in CET is actually a multiple of 5, what is the number of candidates who are at or above the 90th percentile overall and at or above the 80th percentile in both P and M in CET?



BIE uses a different process for selection. If any candidate is appearing in the AET by AIE, BIE considers their AET score for final selection provided the candidate is at or above the 80th percentile in P. Any other candidate at or above the 80th percentile in P in CET, but who is not eligible for the AET, is required to appear in a separate test to be conducted by BIE for being considered for final selection. Altogether, there are 400 candidates this year who are at or above the 80th percentile in P.





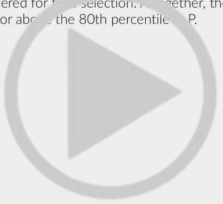
 VIDEO SOLUTION

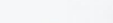
Question 12

If the number of candidates who are at or above the 90th percentile overall and also at or above the 80th percentile in all three sections in CET is actually a multiple of 5, then how many candidates were shortlisted for the AET for AIE?



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 VIDEO SOLUTION



CAT Syllabus PDF



Question 13

If the number of candidates who are at or above the 90th percentile overall and also are at or above the 80th percentile in P in CET, is more than 100, how many candidates had to sit for the separate test for BIE?

- A 299
- B 310
- C 321
- D 330

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BIE uses a different process for selection. If any candidate is appearing in the AET by AIE, BIE considers their AET score for final selection provided the candidate is at or above the 80th percentile in P. Any other candidate at or above the 80th percentile in P in CET, but who is not eligible for the AET, is required to appear in a separate test to be conducted by BIE for being considered for final selection. Together, there are 400 candidates this year who are at or above the 80th percentile in P.



VIDEO SOLUTION

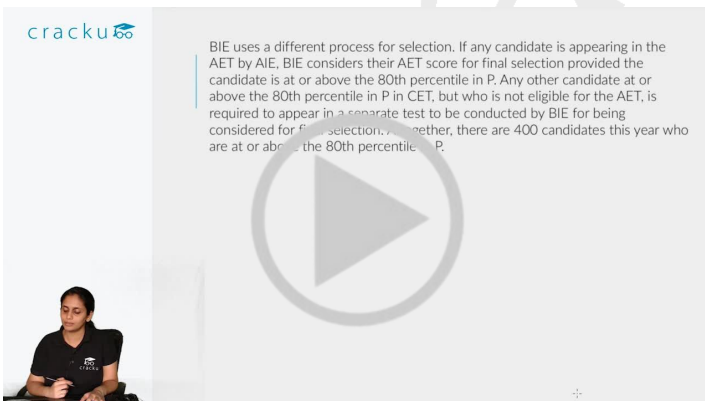
Question 14

What best can be concluded about the number of candidates sitting for the separate test for BIE who were at or above the 90th percentile overall in CET?

- A 3 or 10
- B 10
- C 5
- D 7 or 10

cracku

BIE uses a different process for selection. If any candidate is appearing in the AET by AIE, BIE considers their AET score for final selection provided the candidate is at or above the 80th percentile in P. Any other candidate at or above the 80th percentile in P in CET, but who is not eligible for the AET, is required to appear in a separate test to be conducted by BIE for being considered for final selection. Together, there are 400 candidates this year who are at or above the 80th percentile in P.



VIDEO SOLUTION

Instructions

1600 satellites were sent up by a country for several purposes. The purposes are classified as broadcasting (B), communication (C), surveillance (S), and others (O). A satellite can serve multiple purposes; however a satellite serving either B, or C, or S does not serve O. The following facts are known about the satellites:

1. The numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1.
2. The number of satellites serving all three of B, C, and S is 100.
3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. This number is 30% of the number of satellites exclusively serving B.
4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.

Question 15

If at least 100 of the 1600 satellites were serving O, what can be said about the number of satellites serving S?

- A At most 475
- B Exactly 475
- C No conclusion is possible based on the given information
- D At least 475

The video solution for Question 15 shows a Venn diagram with three overlapping circles labeled B, C, and S. The regions are labeled with letters: B (exclusive), C (exclusive), S (exclusive), BC (exclusive), CS (exclusive), BS (exclusive), and BCS (exclusive). The region outside all three circles is labeled O. The video shows a person solving the problem, with a large play button overlaying the Venn diagram.

VIDEO SOLUTION

The banner for CAT Formula PDF features a person on the left, the text "CAT Formula PDF" in the center, and a PDF icon on the right.

Question 16

If the number of satellites serving at least two among B, C, and S is 1200, which of the following MUST be FALSE?

- A The number of satellites serving B exclusively is exactly 250
- B The number of satellites serving B is more than 1000
- C The number of satellites serving C cannot be uniquely determined
- D All 1600 satellites serve B or C or S

1. The numbers of satellites serving B, C, and S (though they may not exclusively) are in the ratio 2:1:1.
2. The number of satellites serving all three of B, C, and S is 100.
3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. The number is 30% of the number of satellites exclusively serving B.
4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.



Question 17

What is the minimum possible number of satellites serving B exclusively?

- | | |
|----------|-----|
| A | 250 |
| B | 100 |
| C | 500 |
| D | 200 |

A Venn diagram with three overlapping circles labeled B, C, and S. The circles B and C are at the top, overlapping each other, and both overlap with circle S which is positioned below them. The entire diagram is enclosed in a rectangular frame labeled G in the bottom right corner.

1. The numbers of satellites serving B, C, and S (though not exclusively) are in the ratio 2:1:1.
2. The number of satellites serving all three of B, C, and S is 100.
3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. The number is 30% of the number of satellites exclusively serving B.
4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.



Question 18

What best can be said about the number of satellites serving C?

- A** Must be at least 100
- B** Cannot be more than 800
- C** Must be between 450 and 725
- D** Must be between 400 and 800

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1400 satellites were sent up by a country for several purposes. The purposes are classified as broadcasting (B), communication (C), surveillance (S), and others (O). A satellite can serve multiple purposes; however, a satellite serving either B or C or S does not serve O. The following facts are known about the satellites:

1. The numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1.
2. The number of satellites serving all three of B, C, and S is 100.
3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. The number is 30% of the number of satellites exclusively serving B.
4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.

VIDEO SOLUTION

IIM Call Predictor

Accurate analysis of your profile

Instructions

Fun Sports (FS) provides training in three sports - Gilli-danda (G), Kho-Kho (K), and Ludo (L). Currently it has an enrollment of 39 students each of whom is enrolled in at least one of the three sports. The following details are known:

1. The number of students enrolled only in L is double the number of students enrolled in all the three sports.
2. There are a total of 17 students enrolled in G.
3. The number of students enrolled only in G is one less than the number of students enrolled only in L.
4. The number of students enrolled only in K is equal to the number of students who are enrolled in both K and L.
5. The maximum student enrollment is in L.
6. Ten students enrolled in G are also enrolled in at least one more sport.

Question 19

Due to academic pressure, students who were enrolled in all three sports were asked to withdraw from one of the three sports. After the withdrawal, the number of students enrolled in G was six less than the number of students enrolled in L, while the number of students enrolled in K went down by one. After the withdrawal, how many students were enrolled in both G and K?

cracku

Fun Sports (FS) provides training in three sports - Gilli-danda (G), Kho-Kho (K), and Ludo (L). Currently it has an enrollment of 39 students each of whom is enrolled in at least one of the three sports. The following details are known:

1. The number of students enrolled only in L is double the number of students enrolled in all the three sports.
2. There are a total of 17 students enrolled in G.
3. The number of students enrolled only in G is one less than the number of students enrolled only in L.
4. The number of students enrolled only in K is equal to the number of students who are enrolled in both K and L.
5. The maximum student enrollment is in L.
6. Ten students enrolled in G are also enrolled in at least one more sport.

VIDEO SOLUTION

Question 20

Due to academic pressure, students who were enrolled in all three sports were asked to withdraw from one of the three sports. After the withdrawal, the number of students enrolled in G was six less than the number of students enrolled in L, while the number of students enrolled in K went down by one. After the withdrawal, how many students were enrolled in both G and L?

- A 6
- B 5
- C 7
- D 8

cracku

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2. There are a total of 17 students enrolled in G.
3. The number of students enrolled only in G is one less than the number of students enrolled only in L.
4. The number of students enrolled only in K is equal to the number of students who are enrolled in both K and L.
5. The maximum student enrollment is in L.
6. Ten students enrolled in G are also enrolled in at least one more sport.

39

VIDEO SOLUTION

Question 21

If the numbers of students enrolled in K and L are in the ratio 19:22, then what is the number of students enrolled in L?

- A 18
- B 17
- C 22
- D 19

cracku

Fun Sports (FS) provides training in three sports - Gilli-danda (G), Kho-Kho (K), and Ludo (L). Currently it has an enrollment of 39 students each of whom is enrolled in at least one of the three sports. The following details are known:

1. The number of students enrolled only in L is double the number of students enrolled in all the three sports.
2. There are a total of 17 students enrolled in G.
3. The number of students enrolled only in G is one less than the number of students enrolled only in L.
4. The number of students enrolled only in K is equal to the number of students who are enrolled in both K and L.
5. The maximum student enrollment is in L.
6. Ten students enrolled in G are also enrolled in at least one more sport.

39

VIDEO SOLUTION

CAT Percentile Predictor



Question 22

What is the minimum number of students enrolled in both G and L but not in K?

cracku

Fun Sports (FS) provides training in three sports - Gilli-danda (G), Kho-Kho (K), and Ludo (L). Currently it has an enrollment of 39 students each of whom is enrolled in at least one of the three sports. The following details are known:

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2. There are a total of 17 students enrolled in G.
3. The number of students enrolled only in G is one less than the number of students enrolled only in L.
4. The number of students enrolled only in K is equal to the number of students who are enrolled in both K and L.
5. The maximum student enrollment is in L.
6. Ten students enrolled in G are also enrolled in at least one more sport.

Handwritten Venn diagram showing three sets: G (Gilli-danda), K (Kho-Kho), and L (Ludo). The regions are labeled with variables: $2a-1$ for G only, a for K only, $2a$ for L only, a for G and K, a for G and L, a for K and L, and a for all three. The total number of students is 39.

VIDEO SOLUTION

Instructions

There are 15 girls and some boys among the graduating students in a class. They are planning a get-together, which can be either a 1-day event, or a 2-day event, or a 3-day event. There are 6 singers in the class, 4 of them are boys. There are 10 dancers in the class, 4 of them are girls. No dancer in the class is a singer.

Some students are not interested in attending the get-together. Those students who are interested in attending a 3-day event are also interested in attending a 2-day event; those who are interested in attending a 2-day event are also interested in attending a 1-day event.

The following facts are also known:

1. All the girls and 80% of the boys are interested in attending a 1-day event. 60% of the boys are interested in attending a 2-day event.
2. Some of the girls are interested in attending a 1-day event, but not a 2-day event; some of the other girls are interested in attending both.
3. 70% of the boys who are interested in attending a 2-day event are neither singers nor dancers. 60% of the girls who are interested in attending a 2-day event are neither singers nor dancers.
4. No girl is interested in attending a 3-day event. All male singers and 2 of the dancers are interested in attending a 3-day event.
5. The number of singers interested in attending a 2-day event is one more than the number of dancers interested in attending a 2-day event.

Question 23

What BEST can be concluded about the number of male dancers who are interested in attending a 1-day event?

- A 5 or 6
- B 6
- C 5
- D 4 or 6

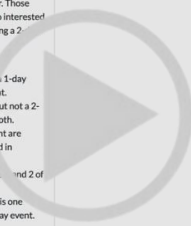
cracku

There are 15 girls and some boys among the graduating students in a class. They are planning a get-together, which can be either a 1-day event, or a 2-day event, or a 3-day event. There are 6 singers in the class, 4 of them are boys. There are 10 dancers in the class, 4 of them are girls. No dancer in the class is a singer.

Some students are not interested in attending the get-together. Those students who are interested in attending a 3-day event are also interested in attending a 2-day event; those who are interested in attending a 2-day event are also interested in attending a 1-day event.

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3. 70% of the boys who are interested in attending a 2-day event are neither singers nor dancers. 60% of the girls who are interested in attending a 2-day event are neither singers nor dancers.
4. No girl is interested in attending a 3-day event. All male singers and 2 of the dancers are interested in attending a 3-day event.
5. The number of singers interested in attending a 2-day event is one more than the number of dancers interested in attending a 2-day event.




 VIDEO SOLUTION

Question 24

How many boys are there in the class?

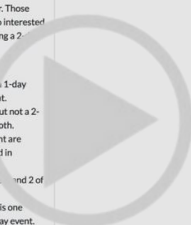
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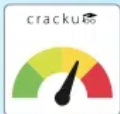
1. All the girls and 80% of the boys are interested in attending a 1-day event. 60% of the boys are interested in attending a 2-day event.
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3. 70% of the boys who are interested in attending a 2-day event are neither singers nor dancers. 60% of the girls who are interested in attending a 2-day event are neither singers nor dancers.
4. No girl is interested in attending a 3-day event. All male singers and 2 of the dancers are interested in attending a 3-day event.
5. The number of singers interested in attending a 2-day event is one more than the number of dancers interested in attending a 2-day event.




 VIDEO SOLUTION



CAT Score Calculator




Question 25

Which of the following can be determined from the given information?

- I. The number of boys who are interested in attending a 1-day event and are neither dancers nor singers.
- II. The number of female dancers who are interested in attending a 1-day event.

- A Only I
- B Neither I nor II
- C Only II
- D Both I and II

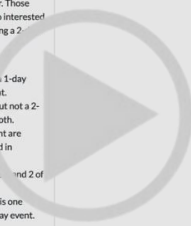
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3. 70% of the boys who are interested in attending a 2-day event are neither singers nor dancers. 60% of the girls who are interested in attending a 2-day event are neither singers nor dancers.
4. No girl is interested in attending a 3-day event. All male singers and 2 of the dancers are interested in attending a 3-day event.
5. The number of singers interested in attending a 2-day event is one more than the number of dancers interested in attending a 2-day event.




 VIDEO SOLUTION

Question 26

How many female dancers are interested in attending a 2-day event?

- A 2
- B 1
- C 0
- D Cannot be determined

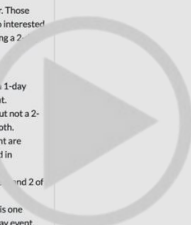
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3. 70% of the boys who are interested in attending a 2-day event are neither singers nor dancers. 60% of the girls who are interested in attending a 2-day event are neither singers nor dancers.
4. No girl is interested in attending a 3-day event. All male singers and 2 of the dancers are interested in attending a 3-day event.
5. The number of singers interested in attending a 2-day event is one more than the number of dancers interested in attending a 2-day event.




 VIDEO SOLUTION

Question 27

What fraction of the class are interested in attending a 2-day event?

- A $\frac{7}{10}$
- B $\frac{7}{13}$
- C $\frac{9}{13}$
- D $\frac{2}{3}$

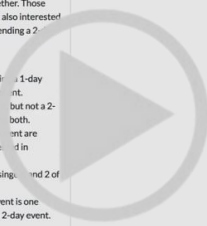
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2. Some of the girls are interested in attending a 1-day event but not a 2-day event; some of the other girls are interested in attending both.
3. 70% of the boys who are interested in attending a 2-day event are neither singers nor dancers. 60% of the girls who are interested in attending a 2-day event are neither singers nor dancers.
4. No girl is interested in attending a 3-day event. All male singers and 2 of the dancers are interested in attending a 3-day event.
5. The number of singers interested in attending a 2-day event is one more than the number of dancers interested in attending a 2-day event.




VIDEO SOLUTION

cracku

CAT Crash Course

Live Classes by IIMA alumni



Instructions

Students in a college are discussing two proposals --

A: a proposal by the authorities to introduce dress code on campus, and

B: a proposal by the students to allow multinational food franchises to set up outlets on college campus.

A student does not necessarily support either of the two proposals.

In an upcoming election for student union president, there are two candidates in fray:

Sunita and Ragini. Every student prefers one of the two candidates.

A survey was conducted among the students by picking a sample of 500 students. The following information was noted from this survey.

1. 250 students supported proposal A and 250 students supported proposal B.
2. Among the 200 students who preferred Sunita as student union president, 80% supported proposal A.
3. Among those who preferred Ragini, 30% supported proposal A.
4. 20% of those who supported proposal B preferred Sunita.
5. 40% of those who did not support proposal B preferred Ragini.
6. Every student who preferred Sunita and supported proposal B also supported proposal A.
7. Among those who preferred Ragini, 20% did not support any of the proposals.

Question 28

How many of the students surveyed supported proposal B, did not support proposal A and preferred Ragini as student union president?

- A 150
- B 210
- C 200
- D 40

cracku

Students in a college are discussing two proposals –
A: a proposal by the authorities to introduce dress code on campus,
 and
B: a proposal by the students to allow multinational food franchises to set up outlets on college campus.

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3. Among those who preferred Ragini, 30% supported proposal A.
4. 20% of those who supported proposal B preferred Sunita.
5. 40% of those who did not support proposal B preferred Ragini.
6. Every student who preferred Sunita and supported proposal B also supported proposal A.
7. Among those who preferred Ragini, 20% did not support any of the proposals.

$A = 250$
 $S = 200$
 $R = 300$
 $T = 500$



 VIDEO SOLUTION

Question 29

What percentage of the students surveyed who supported both proposals A and B preferred Sunita as student union president?

- A 40
- B 25
- C 20
- D 50

cracku

Students in a college are discussing two proposals –
A: a proposal by the authorities to introduce dress code on campus,
 and
B: a proposal by the students to allow multinational food franchises to set up outlets on college campus.

A student does not necessarily support either of the two proposals.

In an upcoming election for student union president, there are two candidates in fray:
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1. 250 students supported proposal A and 250 students supported proposal B.
2. Among the 200 students who preferred Sunita as student union president, 80% supported proposal A.
3. Among those who preferred Ragini, 30% supported proposal A.
4. 20% of those who supported proposal B preferred Sunita.
5. 40% of those who did not support proposal B preferred Ragini.
6. Every student who preferred Sunita and supported proposal B also supported proposal A.
7. Among those who preferred Ragini, 20% did not support any of the proposals.

$A = 250$
 $S = 200$
 $R = 300$
 $T = 500$



 VIDEO SOLUTION

Question 30

What percentage of the students surveyed who did not support proposal A preferred Ragini as student union president?

cracku

Students in a college are discussing two proposals –
A: a proposal by the authorities to introduce dress code on campus,
 and
B: a proposal by the students to allow multinational food franchises to set up outlets on college campus.

A student does not necessarily support either of the two proposals.

In an upcoming election for student union president, there are two candidates in fray:
Sunita and Ragini. Every student prefers one of the two candidates.

A survey was conducted among the students by picking a sample of 500 students. The following information was noted from this survey:

1. 250 students supported proposal A and 250 students supported proposal B.
2. Among the 200 students who preferred Sunita as student union president, 80% supported proposal A.
3. Among those who preferred Ragini, 30% supported proposal A.
4. 20% of those who supported proposal B preferred Sunita.
5. 40% of those who did not support proposal B preferred Sunita.
6. Every student who preferred Sunita and supported proposal B also supported proposal A.
7. Among those who preferred Ragini, 20% did not support any of the proposals.

$A = 250$ $S = 200$
 $T = 300$
 $T = 500$



VIDEO SOLUTION

5000+ CAT Questions

"Cracku CAT Questions"

cracku



Question 31

Among the students surveyed who supported proposal A, what percentage preferred Sunita for student union president?

cracku

Students in a college are discussing two proposals –
A: a proposal by the authorities to introduce dress code on campus,
 and
B: a proposal by the students to allow multinational food franchises to set up outlets on college campus.

A student does not necessarily support either of the two proposals.

In an upcoming election for student union president, there are two candidates in fray:
Sunita and Ragini. Every student prefers one of the two candidates.

A survey was conducted among the students by picking a sample of 500 students. The following information was noted from this survey:

1. 250 students supported proposal A and 250 students supported proposal B.
2. Among the 200 students who preferred Sunita as student union president, 80% supported proposal A.
3. Among those who preferred Ragini, 30% supported proposal A.
4. 20% of those who supported proposal B preferred Sunita.
5. 40% of those who did not support proposal B preferred Sunita.
6. Every student who preferred Sunita and supported proposal B also supported proposal A.
7. Among those who preferred Ragini, 20% did not support any of the proposals.

$A = 250$ $S = 200$
 $T = 300$
 $T = 500$



VIDEO SOLUTION

Instructions

1000 patients currently suffering from a disease were selected to study the effectiveness of treatment of four types of medicines – A, B, C and D. These patients were first randomly assigned into two groups of equal size, called treatment group and control group. The patients in the control group were not treated with any of these medicines; instead they were given a dummy medicine, called placebo, containing only sugar and starch. The following information is known about the patients in the treatment group.

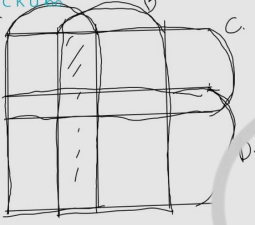
- a. A total of 250 patients were treated with type A medicine and a total of 210 patients were treated with type C medicine.
- b. 25 patients were treated with type A medicine only. 20 patients were treated with type C medicine only. 10 patients were treated with type D medicine only.
- c. 35 patients were treated with type A and type D medicines only. 20 patients were treated with type A and type B medicines only. 30 patients were treated with type A and type C medicines only. 20 patients were treated with type C and type D medicines only.
- d. 100 patients were treated with exactly three types of medicines.
- e. 40 patients were treated with medicines of types A, B and C, but not with medicines of type D. 20 patients

were treated with medicines of types A, C and D, but not with medicines of type B.

f. 50 patients were given all the four types of medicines. 75 patients were treated with exactly one type of medicine.

Question 32

The number of patients who were treated with medicine type D was:



1000 patients currently suffering from a disease were selected to study the effectiveness of treatment of four types of medicines – A, B, C and D. These patients were first randomly assigned into two groups of equal size, called treatment group and control group. The patients in the control group were not treated with any of these medicines; instead they were given a dummy medicine, called placebo, containing only sugar and starch. The following information is known about the treatment group.

a. A total of 250 patients were treated with type A medicine and a total of 210 patients were treated with type C medicine.

b. 25 patients were treated with type A medicine only, 20 patients were treated with type C medicine only, 10 patients were treated with type D medicine only.

c. 40 patients were treated with type A and type D medicines only, 20 patients were treated with type C and type D medicines only, 30 patients were treated with type A and type C medicines only.

d. 40 patients were treated with exactly three types of medicines.

e. 40 patients were treated with medicines of types A, B and C, but not with medicines of type D. 50 patients were treated with medicines of types A, C and D, but not with medicine type B.

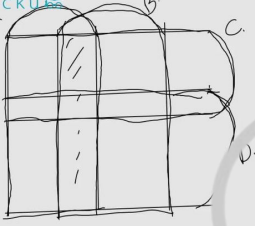
f. 50 patients were given all the four types of medicines. 75 patients were treated with exactly one type of medicine.



VIDEO SOLUTION

Question 33

How many patients were treated with medicine type B?



1000 patients currently suffering from a disease were selected to study the effectiveness of treatment of four types of medicines – A, B, C and D. These patients were first randomly assigned into two groups of equal size, called treatment group and control group. The patients in the control group were not treated with any of these medicines; instead they were given a dummy medicine, called placebo, containing only sugar and starch. The following information is known about the treatment group.

a. A total of 250 patients were treated with type A medicine and a total of 210 patients were treated with type C medicine.

b. 25 patients were treated with type A medicine only, 20 patients were treated with type C medicine only, 10 patients were treated with type D medicine only.

c. 40 patients were treated with type A and type D medicines only, 20 patients were treated with type C and type D medicines only, 30 patients were treated with type A and type C medicines only.

d. 40 patients were treated with exactly three types of medicines.

e. 40 patients were treated with medicines of types A, B and C, but not with medicines of type D. 50 patients were treated with medicines of types A, C and D, but not with medicine type B.

f. 50 patients were given all the four types of medicines. 75 patients were treated with exactly one type of medicine.



VIDEO SOLUTION



CAT Daily Targets

Step by step, heading towards 99+ percentile



Question 34

The number of patients who were treated with medicine types B, C and D, but not type A was:

cracku

1000 patients currently suffering from a disease were selected to study the effectiveness of treatment of four types of medicines – A, B, C and D. These patients were first randomly assigned into two groups of equal size, called treatment group and control group. The patients in the control group were not treated with any of these medicines; instead they were given a dummy medicine, called placebo, containing only sugar and starch. The following information is known about the treatment group.

a. A total of 250 patients were treated with type A medicine and a total of 210 patients were treated with type C medicine.
 b. 25 patients were treated with type A medicine only, 20 patients were treated with type C medicine only, 10 patients were treated with type D medicine only.
 c. 40 patients were treated with type A and type D medicines only, 20 patients were treated with type C and type D medicines only, 30 patients were treated with type A and type D medicines only.
 d. 40 patients were treated with medicines of types A, B and C, but not with medicines of type D. 20 patients were treated with medicines of types A, C and D, but not with medicines of type B.
 e. 50 patients were given all the four types of medicines. 75 patients were treated with only one type of medicine.

VIDEO SOLUTION

Question 35

How many patients were treated with medicine types B and D only?

cracku

1000 patients currently suffering from a disease were selected to study the effectiveness of treatment of four types of medicines – A, B, C and D. These patients were first randomly assigned into two groups of equal size, called treatment group and control group. The patients in the control group were not treated with any of these medicines; instead they were given a dummy medicine, called placebo, containing only sugar and starch. The following information is known about the treatment group.

a. A total of 250 patients were treated with type A medicine and a total of 210 patients were treated with type C medicine.
 b. 25 patients were treated with type A medicine only, 20 patients were treated with type C medicine only, 10 patients were treated with type D medicine only.
 c. 40 patients were treated with type A and type D medicines only, 20 patients were treated with type C and type D medicines only, 30 patients were treated with type A and type D medicines only.
 d. 40 patients were treated with medicines of types A, B and C, but not with medicines of type D. 20 patients were treated with medicines of types A, C and D, but not with medicines of type B.
 e. 50 patients were given all the four types of medicines. 75 patients were treated with only one type of medicine.

VIDEO SOLUTION

Instructions

Ten musicians (A, B, C, D, E, F, G, H, I and J) are experts in at least one of the following three percussion instruments: tabla, mridangam, and ghatam. Among them, three are experts in tabla but not in mridangam or ghatam, another three are experts in mridangam but not in tabla or ghatam, and one is an expert in ghatam but not in tabla or mridangam. Further, two are experts in tabla and mridangam but not in ghatam, and one is an expert in tabla and ghatam but not in mridangam.

The following facts are known about these ten musicians.

- Both A and B are experts in mridangam, but only one of them is also an expert in tabla.
- D is an expert in both tabla and ghatam.
- Both F and G are experts in tabla, but only one of them is also an expert in mridangam.
- Neither I nor J is an expert in tabla.
- Neither H nor I is an expert in mridangam, but only one of them is an expert in ghatam.

Question 36

Who among the following is DEFINITELY an expert in tabla but not in either mridangam or ghatam?

- A F
- B C
- C A

D H

The video solution for Question 37 displays a Venn diagram with three overlapping circles labeled 'Tabla', 'Mrid.', and 'Ghatam'. The 'Tabla' circle contains the number 3, the 'Mrid.' circle contains 2, and the 'Ghatam' circle contains 1. The intersection of 'Tabla' and 'Mrid.' contains 2, the intersection of 'Tabla' and 'Ghatam' contains 1, and the intersection of 'Mrid.' and 'Ghatam' contains 1. The intersection of all three circles contains 1. The video also shows a list of facts and a list of options.

The following facts are known about these ten musicians.

1. Both A and B are experts in mridangam, but only one of them is also an expert in tabla.
2. D is an expert in both tabla and ghatam.
3. Both F and G are experts in tabla, but only one of them is also an expert in mridangam.
4. Neither I nor J is an expert in tabla.
5. Neither H nor I is an expert in mridangam, but only one of them is an expert in ghatam.

Options: A, B, C, D, E, F, G, H, I, J

VIDEO SOLUTION

CAT Sectional Tests

Take Here

Question 37

Which of the following pairs CANNOT have any musician who is an expert in both tabla and mridangam but not in ghatam?

- A F and G
- B C and E
- C A and B
- D C and F

The video solution for Question 37 displays a Venn diagram with three overlapping circles labeled 'Tabla', 'Mrid.', and 'Ghatam'. The 'Tabla' circle contains the number 3, the 'Mrid.' circle contains 2, and the 'Ghatam' circle contains 1. The intersection of 'Tabla' and 'Mrid.' contains 2, the intersection of 'Tabla' and 'Ghatam' contains 1, and the intersection of 'Mrid.' and 'Ghatam' contains 1. The intersection of all three circles contains 1. The video also shows a list of facts and a list of options.

The following facts are known about these ten musicians.

1. Both A and B are experts in mridangam, but only one of them is also an expert in tabla.
2. D is an expert in both tabla and ghatam.
3. Both F and G are experts in tabla, but only one of them is also an expert in mridangam.
4. Neither I nor J is an expert in tabla.
5. Neither H nor I is an expert in mridangam, but only one of them is an expert in ghatam.

Options: A, B, C, D, E, F, G, H, I, J

VIDEO SOLUTION

Question 38

If C is an expert in mridangam and F is not, then which are the three musicians who are experts in tabla but not in either mridangam or ghatam?

- A E, F and H

The following facts are also known:

1. There were equal numbers of domestic and foreign products.
2. Half of the domestic products were FDA approved cosmetic products.
3. None of the foreign products had both the approvals, while 60 domestic products had both the approvals.
4. There were 140 nutrition products, half of them were foreign products.
5. There were 200 FDA approved products. 70 of them were foreign products and 120 of them were cosmetic products.

Question 40

How many foreign products were FDA approved cosmetic products?

A speciality supermarket sells 320 products. Each of these products was either a cosmetic product or a nutrition product. Each of these products was also either a foreign product or a domestic product. Each of these products had at least one of the two approvals - FDA or EU.

The following facts are also known:

1. There were equal numbers of domestic and foreign products.
2. Half of the domestic products were FDA approved cosmetic products.
3. None of the foreign products had both the approvals, while 60 domestic products had both the approvals.
4. There were 140 nutrition products, half of them were foreign products.
5. There were 200 FDA approved products. 70 of them were foreign products and 120 of them were cosmetic products.

The video shows a hand-drawn Venn diagram with two main sets: C (Cosmetic) and N (Nutrition). C is divided into FDA (120) and EU (160). N is divided into FDA (70) and EU (140). The intersection of C and N is FDA (60). The intersection of C and N is EU (100). The intersection of C and N is FDA (60).

VIDEO SOLUTION

Question 41

How many cosmetic products did not have FDA approval?

- A 10
- B Cannot be determined
- C 50
- D 60

A speciality supermarket sells 320 products. Each of these products was either a cosmetic product or a nutrition product. Each of these products was also either a foreign product or a domestic product. Each of these products had at least one of the two approvals - FDA or EU.

The following facts are also known:

1. There were equal numbers of domestic and foreign products.
2. Half of the domestic products were FDA approved cosmetic products.
3. None of the foreign products had both the approvals, while 60 domestic products had both the approvals.
4. There were 140 nutrition products, half of them were foreign products.
5. There were 200 FDA approved products. 70 of them were foreign products and 120 of them were cosmetic products.

The video shows a hand-drawn Venn diagram with two main sets: C (Cosmetic) and N (Nutrition). C is divided into FDA (120) and EU (160). N is divided into FDA (70) and EU (140). The intersection of C and N is FDA (60). The intersection of C and N is EU (100). The intersection of C and N is FDA (60).

VIDEO SOLUTION

Question 42

If 70 cosmetic products did not have EU approval, then how many nutrition products had both the approvals?

- A 50

B 30

C 10

D 20



A speciality supermarket sells 320 products. Each of these products was either a cosmetic product or a nutrition product. Each of these products was also either a foreign product or a domestic product. Each of these products had at least one of the two approvals - FDA or EU.

The following facts are also known:

1. There were equal numbers of domestic and foreign products.
2. Half of the domestic products were FDA approved cosmetic products.
3. None of the foreign products had both the approvals, while 60 domestic products had both the approvals.
4. There were 140 nutrition products, half of them were foreign products.
5. There were 200 FDA approved products. 70 of them were foreign products and 130 were domestic products.

Handwritten solution diagram:

```
graph TD
    Total[320] --> C[Cosmetic]
    Total --> N[Nutrition]
    C --> F[Cosmetic Foreign]
    C --> D[Cosmetic Domestic]
    N --> F[Nutrition Foreign]
    N --> D[Nutrition Domestic]
    F --> FDA[140]
    F --> EU[140]
    D --> FDA[130]
    D --> EU[130]
    FDA --> FDA_EU[130]
    FDA --> FDA_N[100]
    EU --> FDA_EU[130]
    EU --> EU_N[100]
    FDA_EU --> FDA_EU_C[60]
    FDA_EU --> FDA_EU_N[70]
    FDA_N --> FDA_N_C[10]
    FDA_N --> FDA_N_N[90]
    EU_N --> EU_N_C[10]
    EU_N --> EU_N_N[90]
```

VIDEO SOLUTION



Question 43

If 50 nutrition products did not have EU approval, then how many domestic cosmetic products did not have EU approval?



A speciality supermarket sells 320 products. Each of these products was either a cosmetic product or a nutrition product. Each of these products was also either a foreign product or a domestic product. Each of these products had at least one of the two approvals - FDA or EU.

The following facts are also known:

1. There were equal numbers of domestic and foreign products.
2. Half of the domestic products were FDA approved cosmetic products.
3. None of the foreign products had both the approvals, while 60 domestic products had both the approvals.
4. There were 140 nutrition products, half of them were foreign products.
5. There were 200 FDA approved products. 70 of them were foreign products and 130 were domestic products.

Handwritten solution diagram:

```
graph TD
    Total[320] --> C[Cosmetic]
    Total --> N[Nutrition]
    C --> F[Cosmetic Foreign]
    C --> D[Cosmetic Domestic]
    N --> F[Nutrition Foreign]
    N --> D[Nutrition Domestic]
    F --> FDA[140]
    F --> EU[140]
    D --> FDA[130]
    D --> EU[130]
    FDA --> FDA_EU[130]
    FDA --> FDA_N[100]
    EU --> FDA_EU[130]
    EU --> EU_N[100]
    FDA_EU --> FDA_EU_C[60]
    FDA_EU --> FDA_EU_N[70]
    FDA_N --> FDA_N_C[10]
    FDA_N --> FDA_N_N[90]
    EU_N --> EU_N_C[10]
    EU_N --> EU_N_N[90]
```

VIDEO SOLUTION

Question 44

Which among the following options best represents the number of domestic cosmetic products that had both the approvals?

A At least 10 and at most 60

B At least 10 and at most 80

C At least 20 and at most 70

D At least 20 and at most 50

320 = C

FDA/EU
F
(160)

C
N

N
D
(160)
N

C
FDA(80)

A speciality supermarket sells 320 products. Each of these products was either a cosmetic product or a nutrition product. Each of these products was also either a foreign product or a domestic product. Each of these products had at least one of the two approvals - FDA or EU.

The following facts are also known:

1. There were equal numbers of domestic and foreign products.
2. Half of the domestic products were FDA approved cosmetic products.
3. None of the foreign products had both the approvals, while 60 domestic products had both the approvals.

There were 140 nutrition products, half of them were foreign products, and 200 FDA approved cosmetic products. 70 of them were foreign products and 130 were domestic products.

 VIDEO SOLUTION

Instructions

A survey of 600 schools in India was conducted to gather information about their online teaching learning processes (OTLP). The following four facilities were studied.

F1: Own software for OTLP

F2: Trained teachers for OTLP

F3: Training materials for OTLP

F4: All students having Laptops

The following observations were summarized from the survey.

1. 80 schools did not have any of the four facilities - F1, F2, F3, F4.
2. 40 schools had all four facilities.
3. The number of schools with only F1, only F2, only F3, and only F4 was 25, 30, 26 and 20 respectively.
4. The number of schools with exactly three of the facilities was the same irrespective of which three were considered.
5. 313 schools had F2.
6. 26 schools had only F2 and F3 (but neither F1 nor F4).
7. Among the schools having F4, 24 had only F3, and 45 had only F2.
8. 162 schools had both F1 and F2.
9. The number of schools having F1 was the same as the number of schools having F4.

Question 45

What was the number of schools having only facilities F1 and F4?

A survey of 600 schools in India was conducted to gather information about their online teaching learning processes (OTLP). The following four facilities were studied.

E1: Own software for OTLP
F2: Trained teachers for OTLP
F3: Computer materials for OTLP
F4: All students having Laptops

The following observations were summarized from the survey.

520 schools had all four of the facilities - F1, F2, F3, F4.
 380 schools had only F1, only F2, only F3, and only F4 was 25.
 320 schools had exactly two of the facilities - F1 and F2, F1 and F3, and F2 and F3.
 120 schools had exactly one of the facilities - F1, F2, or F3.
 100 schools had none of the facilities.

5. 313 schools had F1 and F2 but neither F3 nor F4.
 7. Among the schools having F4, 24 had only F3, and 45 had only F2.
 8. Among the schools having F1, 24 had both F1 and F2.
 9. The number of schools having F1 was the same as the number of schools having F4.

 VIDEO SOLUTION

100+ CAT Quant Questions



Question 46

What was the number of schools having facilities F2 and F4?

- A 185
- B 95
- C 45
- D 85

A survey of 600 schools in India was conducted to gather information about their online teaching learning processes (OTLP). The following four facilities were studied.

E1: Own software for OTLP
F2: Trained teachers for OTLP
F3: Materials for OTLP
F4: All students having Laptops

The following observations were summarized from the survey:

- 10 schools did not have any of the four facilities - F1, F2, F3, F4.
- 25 schools had all four facilities.
- 25 schools had only F1, only F2, only F3, and only F4 was 25.
- 5 schools had exactly two of the facilities.
- 5 schools had exactly three of the facilities was the same.
- 5 schools had F1.
- 26 schools had F2 and F3 (but neither F1 nor F4).
7. Among the schools having F4, 24 had only F3, and 45 had only F2.
8. 45 schools had both F1 and F2.
9. The number of schools having F1 was the same as the number of schools having F4.

VIDEO SOLUTION

Question 47

What was the total number of schools having exactly three of the four facilities?

- A 64
- B 50
- C 200
- D 80

A survey of 600 schools in India was conducted to gather information about their online teaching learning processes (OTLP). The following four facilities were studied.

E1: Own software for OTLP
F2: Trained teachers for OTLP
F3: Materials for OTLP
F4: All students having Laptops

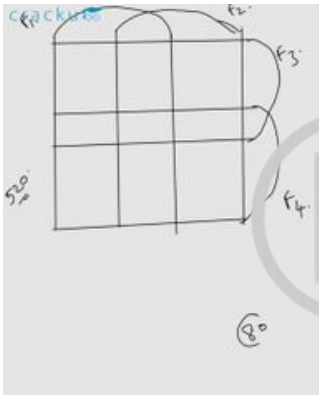
The following observations were summarized from the survey:

- 10 schools did not have any of the four facilities - F1, F2, F3, F4.
- 25 schools had all four facilities.
- 25 schools had only F1, only F2, only F3, and only F4 was 25.
- 5 schools had exactly two of the facilities.
- 5 schools had exactly three of the facilities was the same.
- 5 schools had F1.
- 26 schools had F2 and F3 (but neither F1 nor F4).
7. Among the schools having F4, 24 had only F3, and 45 had only F2.
8. 45 schools had both F1 and F2.
9. The number of schools having F1 was the same as the number of schools having F4.

VIDEO SOLUTION

Question 48

What was the number of schools having only facilities F1 and F3?



A survey of 600 schools in India was conducted to gather information about their online teaching-learning processes (OTLP). The following four facilities were studied:

- F1: Own software for OTLP
- F2: Trained teachers for OTLP
- F3: Trained materials for OTLP
- F4: All students having Laptops

The following observations were summarized from the survey:

- 50 schools did not have any of the four facilities - F1, F2, F3, F4.
- 80 schools had all four facilities.
- 25 schools had only F1, only F2, only F3, and only F4 was 25.
- The number of schools with exactly three of the facilities was the same as the number of schools with exactly two of the facilities.
- 5, 313 schools had F1 and F2 (but neither F3 nor F4).
- 25 schools had F2 and F3 (but neither F1 nor F4).
- Among the schools having F4, 24 had only F3, and 45 had only F2.
- 25 schools had both F1 and F2.
- The number of schools having F1 was the same as the number of schools having F4.

VIDEO SOLUTION

100+ CAT DILR Questions

